# State machines for bounded agents

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for State machines for bounded agents before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Testing pure components

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported State machines for bounded agents trace and record the first step you can explain without help.

Use W2: Trace the worked case for **State** machines for bounded agents. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed State machines for bounded agents explanation against the supplied facts before discussing it.

Use the worked case for **State** machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Changed independent case

Only these **transition**s are allowed: ready→looking_up→review→done. Done is terminal. Evaluate ready→review→done and ready→looking_up→review→done.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied State machines for bounded agents facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied State machines for bounded agents facts and keep your reasoning.

Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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WORKBOOK · 7 OF 15

## W3 Audit the limit

> Audit the limit. Apply the supplied State machines for bounded agents facts and keep your reasoning.

A report omits this qualification: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied State machines for bounded agents facts and keep your reasoning.

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid **transition**.

**My reasoning and evidence**

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## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied State machines for bounded agents facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

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WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect State machines for bounded agents to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **state** machines for bounded agents **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **state** machines for bounded agents.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

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WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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## Pictorial case practice: State machines for bounded agents

> A parcel lookup state machine. An agent may follow idle→querying→checking→complete; complete is terminal. Explain the labelled step before checking the changed condition.

A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal.

A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

AI ACADEMY / AI-6/13

# State machines for bounded agents

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



**1. Read arrows**

The graph permits three forward transitions.



**2. Follow A**

All adjacent pairs match permitted arrows.



**3. Inspect B**

Querying -> complete skips checking and is invalid.

**CHANGE ONE THING**

Add checking -> querying as a permitted retry.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Querying→complete skips checking and is invalid.

1. Follow A

2. Read arrows

3. Inspect B

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect B Querying→complete skips checking and is invalid. A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Add checking→querying as a permitted retry. Which conclusion is supported by the supplied facts?

1. That retry becomes valid; querying→complete stays forbidden.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: That retry becomes valid; querying→complete stays forbidden. Adding one arrow does not authorise unrelated shortcuts.

**Repair a missing building block**

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

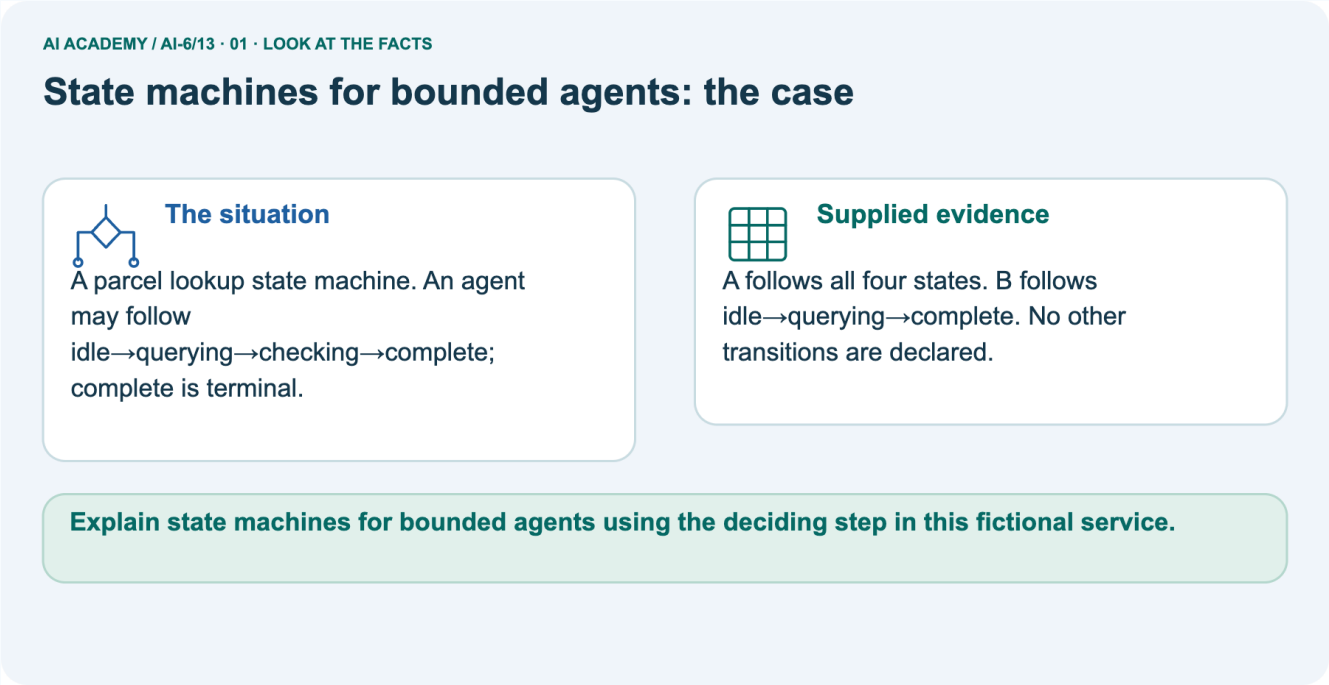
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## Visual concept journey

> Explain each State machines for bounded agents picture from its labels, then record a relevant limit.

## 01. State machines for bounded agents: the case



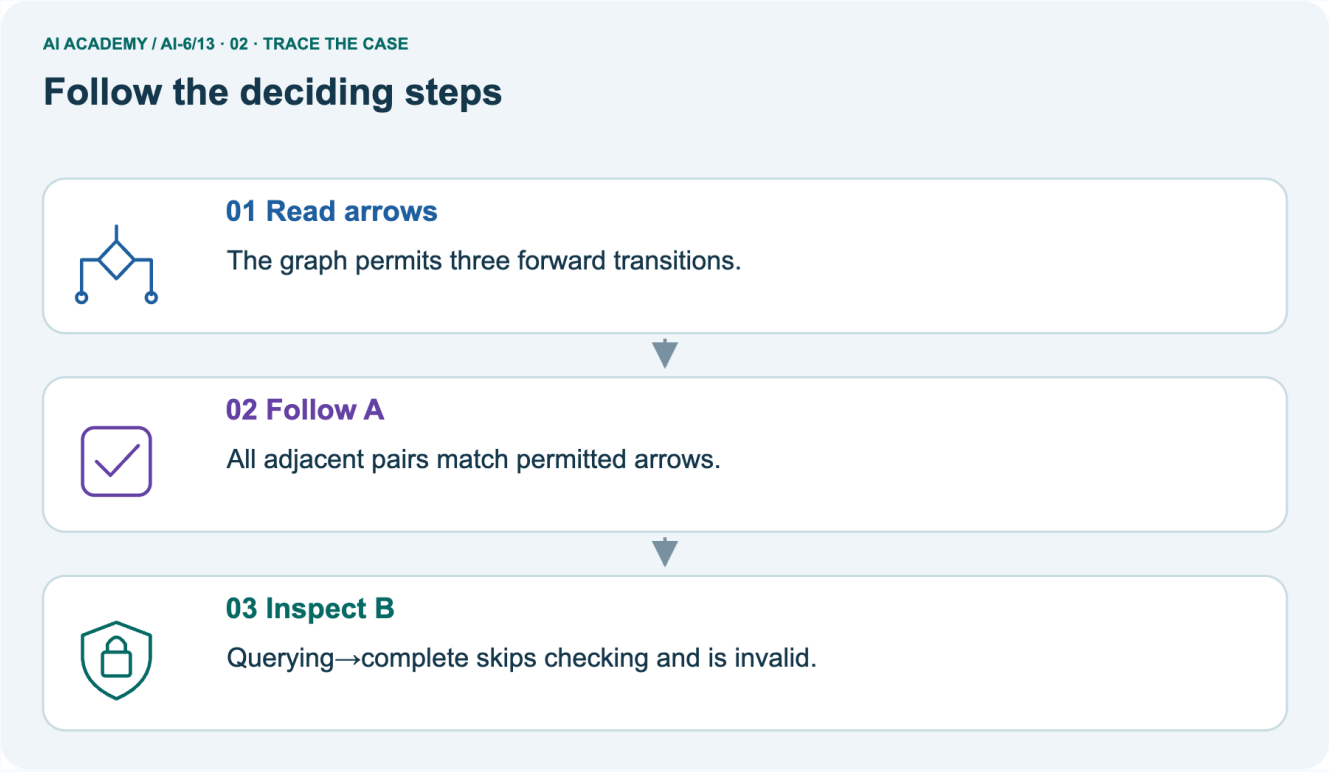

Follow the labelled picture with the matching explanation. Open the vector diagram

A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal. Supplied facts: A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



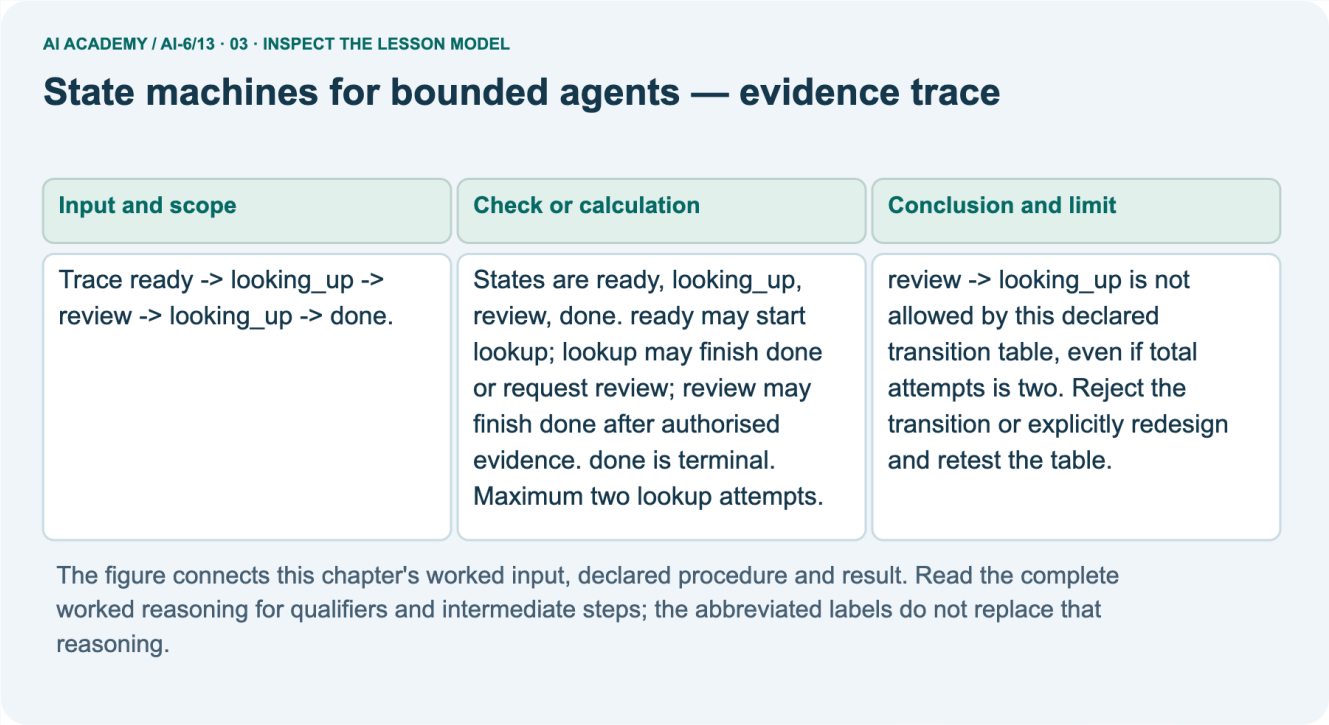

Follow the labelled picture with the matching explanation. Open the vector diagram

Read arrows: The graph permits three forward **transition**s. → Follow A: All adjacent pairs match permitted arrows. → Inspect B: Querying→complete skips checking and is invalid.

> Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

### 03. State machines for bounded agents — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

**State** machines for bounded agents — **evidence** trace.

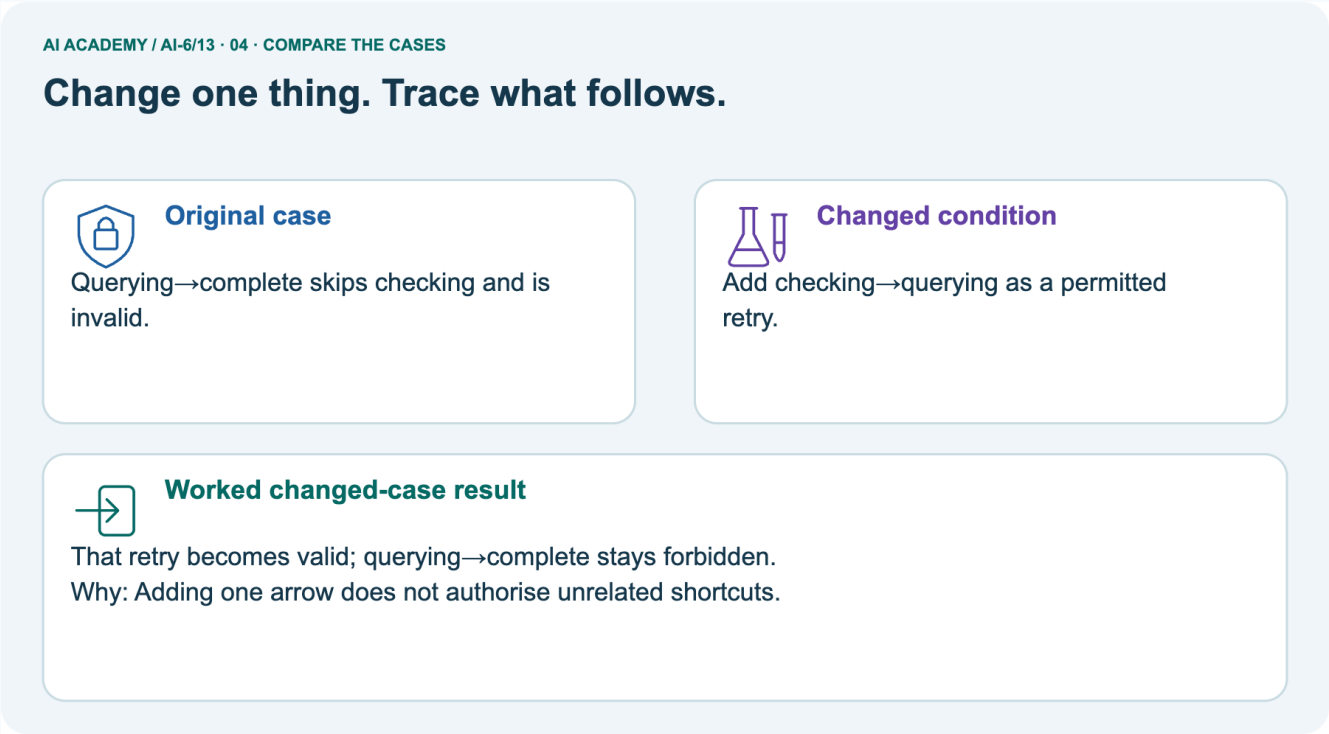
Trace ready -> looking_up -> review -> looking_up -> done. | **State**s are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal.

Maximum two lookup attempts. | review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

> Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

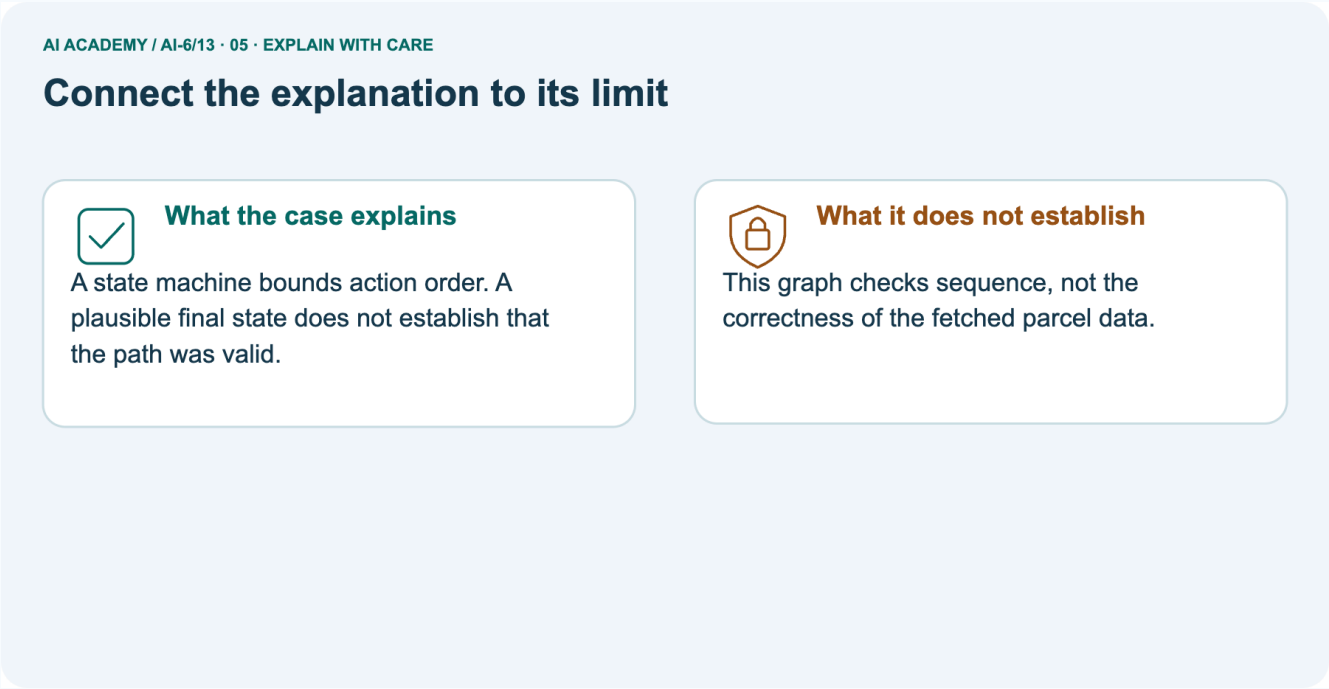
Original case: Querying→complete skips checking and is invalid. Changed condition: Add checking→querying as a permitted retry. Expected result: That retry becomes valid; querying→complete stays forbidden.

Reason: Adding one arrow does not authorise unrelated shortcuts.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit



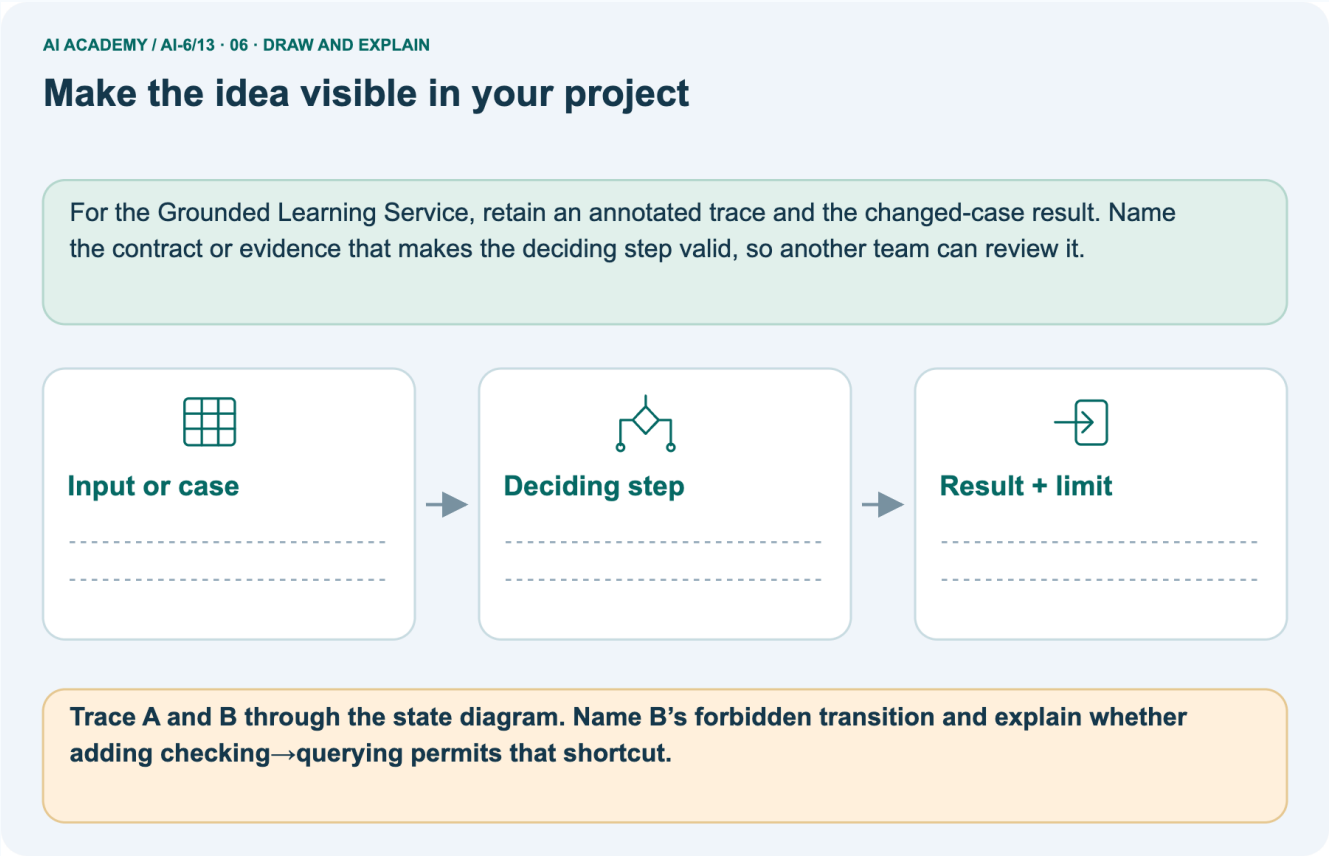

Follow the labelled picture with the matching explanation. Open the vector diagram

A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid. Limit: This graph checks sequence, not the correctness of the fetched parcel data.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.